

OpenTelemetry

... Playtime is Over



Josh Lee (Altinity)
Adriana Villela (Dynatrace)



Josh Lee
Open Source
Advocate @ Altinity

Altinity® is a Registered Trademark of Altinity, Inc. ClickHouse® is a registered trademark of ClickHouse, Inc.; Altinity is not affiliated with or associated with ClickHouse, Inc. We are but humble open source contributors.

01

We're past the
OTel
honeymoon

Time to grow up



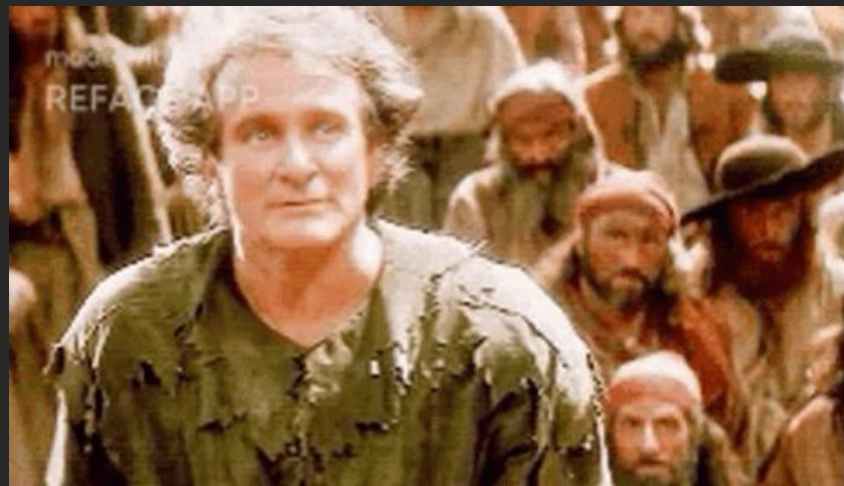
Managing costs



Telemetry quality



Collector fleet management



Don't Be Tempted by...



Auto-instrumentation
overload



Tracing as an afterthought



Instrumentation outsourcing



Letting AI take the wheel



**The easy path
isn't always the
best path**

02

Cutting down the noise (and costs)

PROBLEM:

Information overload

Intentional instrumentation



Built-in is best 😎



Library instrumentation: 👍



No-code instrumentation: 😐



Standards: Weaver, MCP 😊



Auto-instrumentation gotchas



Lots of spans



Can be configured*



Computational overhead*



Filter as soon as possible



Sampling



We don't care about every
HTTP 200 response



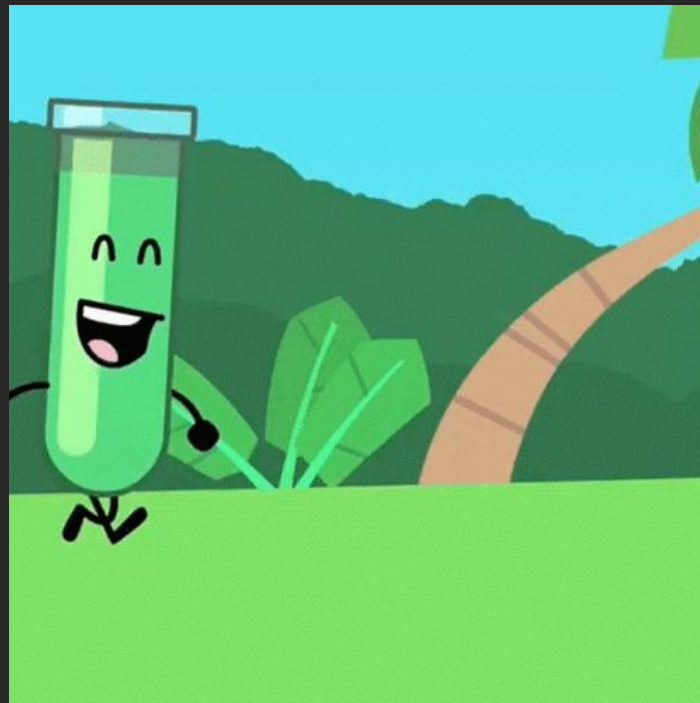
Reduce costs without losing
visibility



Filter first, then sample



Head vs Tail sampling



Metrics



Metrics are deceptively complex



Cardinality, schmardinality



Don't scrape what you don't need



Target allocator is 🙌



03

Pipeline efficiency

PROBLEM:
Telemetry is
expensive!



OTLP protobuf ain't perfect



Hierarchical telemetry data
Repetitive metadata



OTLP is row-based
Inefficient for storage



Serialize/deserialize
Encoding/decoding gets expensive



CNCF + Apache = OTel Arrow



OTel Arrow is based on the Apache Arrow framework, and serves both a data format and a transport protocol.



OTel Arrow



Columnar format



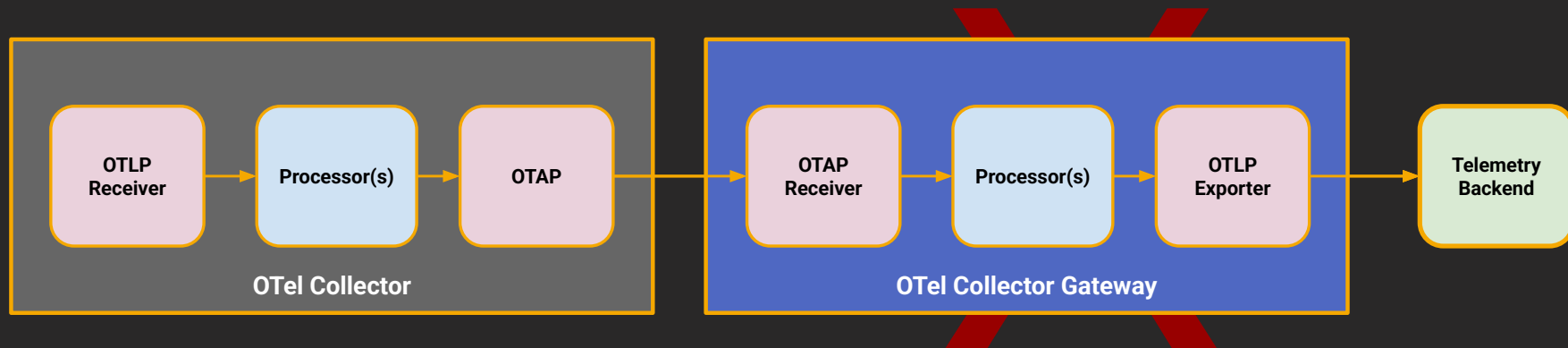
Zero-copy data sharing



Available as Collector components



Current State



Next phase

Telemetry
Data
Source



Telemetry
Backend

04

Schema management & validation

PROBLEM:

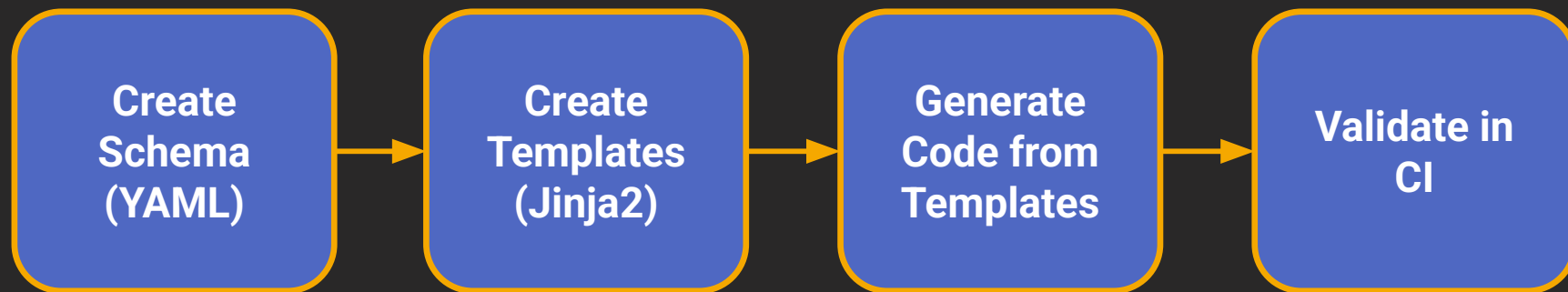
Managing semantic convention drift

OTel Weaver to the Rescue!

A tool for creating, documenting, managing, and validating OpenTelemetry schemas and policies.



How it works



Requires learning Rego & Jinja2

Weaver
Overview



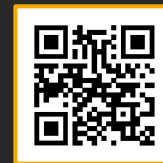
05

Collector Management

PROBLEM:
Oh so many
Collectors to
configure

OpAMP to the rescue!

Remote management of
data collection agents!



↑
OpAMP
Overview

OpAMP client & server



OpAMP Server

Sends config updates, packages, instructions



OpAMP Client

Reports status, applies updates, receives configs

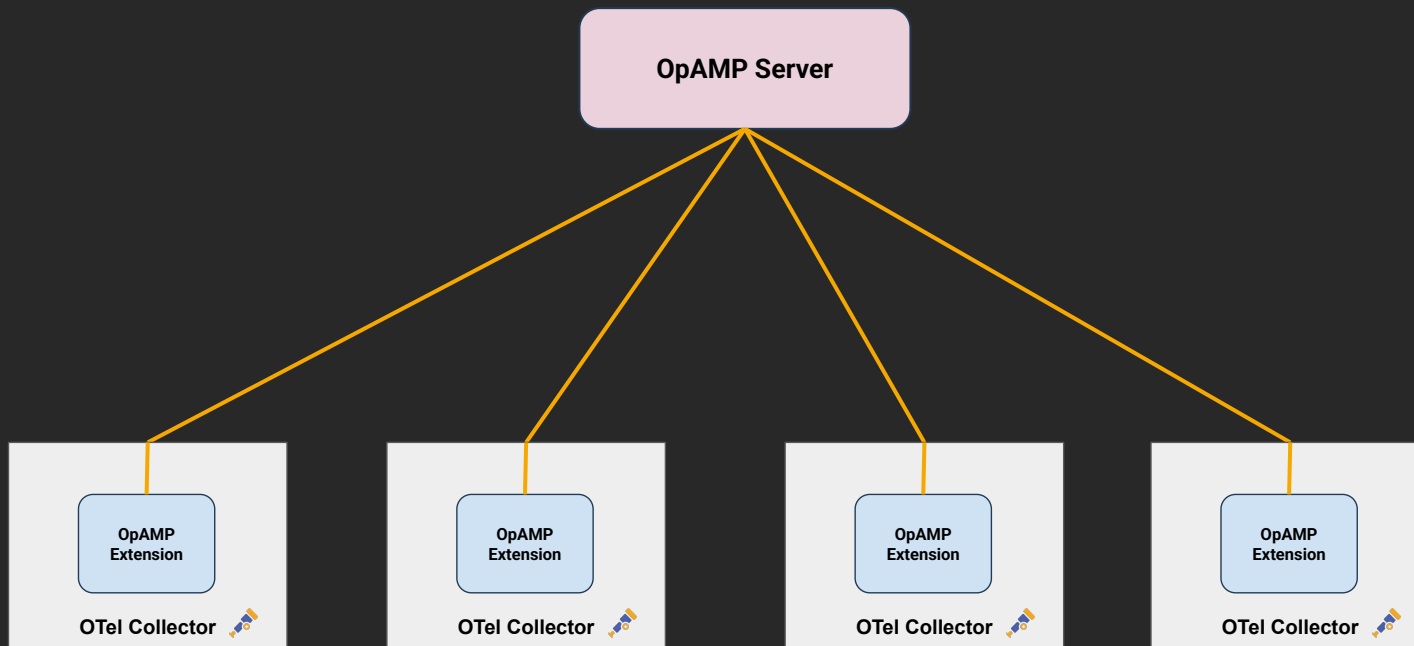


Choose from existing server or write your own!

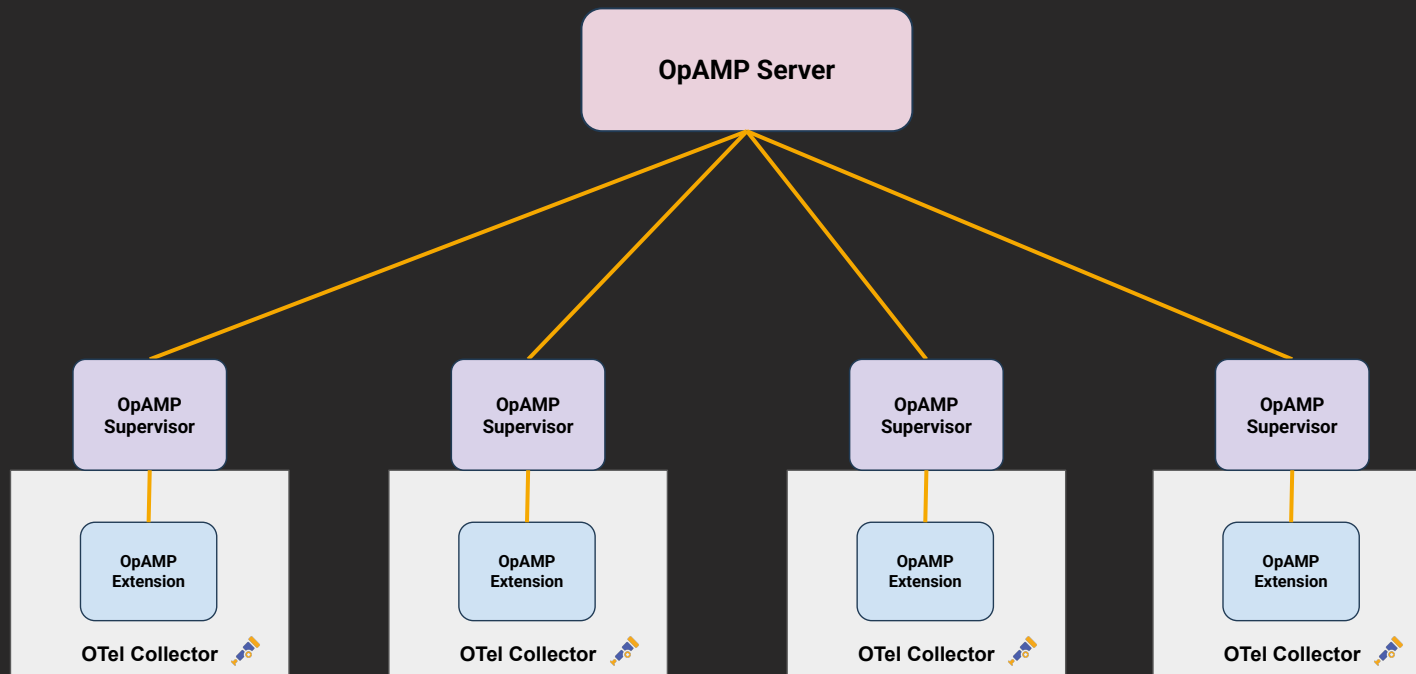


OpAMP on the Collector?

OpAMP Extension



OpAMP Supervisor



OTel tools gotchas



Many moving parts



Large learning curve



Sparsely documented



06

Now WHAT?

OTel Maturity Matrix (#goals)

Basics	Good	Better	At Scale
• RED Metrics • Resource utilization • Logging with levels • Black-box testing	• Tracing for key paths • Monitoring for your monitoring • Custom dimensions / tags • SLOs	• Prevalent tracing • Instrumented internal libraries • Internal sem convs and docs with Weaver	• Optimized pipelines with Arrow • Fleet management with OpAMP • Weaver live-checking

What you can do NOW!



Look at areas for improvement



Use AI agents to stay up to date



Don't boil the ocean



Ask for help



Don't panic!



Thank You!

🦋 @joshleecreates.bsky.social

🐘 @joshleecreates@hachyderm.io

🧳 linkedin.com/in/joshuamlee

💬 altinity.com/slack

Connect w/ me



Thank You!

 @joshleecreates.bsky.social

 @joshleecreates@hachyderm.io

 linkedin.com/in/joshuamlee

 altinity.com/slack

Resources & Slides



BONUS:

Collector-ing “well”

BYOC (Build Your Own Collector)



Most people use Collector contrib



Contrib == bloat + insecure



Build your own Collector



Use OTel Collector builder



Collector Deployment Patterns



Gateway(s) + sidecars



Collector per signal



StatefulSet



No Collector (serverless)



Audits



Coverage



Quality



Usefulness



Pipeline Health



Governance

